

SHADE TOLERANCE IN TREE SEEDLINGS

II. GROWTH ANALYSIS OF PLANTS RAISED UNDER ARTIFICIAL SHADE

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SUMMARY

Seedlings of five tree species of different shade tolerance were raised in 100, 44, 17 and 3% of natural daylight. Growth analysis was carried out on data obtained from two harvests during the growing season. The net assimilation rates (E) of the more tolerant species were little higher in 100% daylight than in slight shade (44% daylight), but thereafter E decreased as shade increased. The intolerant species had greatest E in 100% daylight, but this high E fell sharply with increasing shade, and in *Populus tremuloides* was negative in 3% daylight. Even in this deep shade, however, the intolerant *Liriodendron tulipifera* had as high a net assimilation rate as the shade-tolerant species. Decreased E was offset by increased leaf area ratio (leaf area/plant weight), but in contrast to previous studies with herbaceous species, the magnitude of this increase gave no indication of the relative shade tolerance of the different species. In *L. tulipifera* and *Acer rubrum*, the increased leaf area ratio resulted from an increase in the proportion of assimilates diverted into leaf production (i.e. increased leaf weight ratio), together with a relatively greater increase in specific leaf area (leaf area/leaf weight). The more tolerant *Fagus grandifolia* and *Quercus rubra* showed less increase in specific leaf area, and leaf weight ratio increased only in the deepest shade. No clear-cut distinction can be drawn between tolerant and intolerant species on the basis of the relative size or efficiency of their photosynthetic systems in low light; factors other than those relating to carbohydrate economy in shade, also play a role.

INTRODUCTION

Continued growth and survival in shade implies a net gain in carbohydrates by the plant. Differences between species in their ability to tolerate shade may therefore be explained by their relative efficiencies in photosynthesis at low light intensity, and limitation of losses through respiration. In the first paper of this series (Loach, 1967), these differences were examined in five north-east American tree species.

Shade also influences plant morphology, and changes in the size of the photosynthetic system must be considered alongside changes in its efficiency. Investigations of herbaceous species have indicated that the effects of shade on gross morphology may be at least as important as changes in leaf physiology (Blackman and Wilson, 1951b; Blackman and Black, 1959). Comparatively few similar studies have been made with woody plants. Here, growth analysis is used to assess the relative importance of changes in the efficiency and size of the photosynthetic system.

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Table 1. *The species used, timing of the experiments and meteorological data*

Species	Tolerance	Age when assigned to treatment (days)	Days pre-adaptation before 1st harvest	Days between harvests	Harvest dates <i>t</i> ₁ <i>t</i> ₂	Replication*	Mean hours sunshine/day	Average temperature (°C)†
1965								
<i>Fagus grandifolia</i>	Very tolerant	41	31	62	9 July-9 Sept.	9	9.5	71
<i>Acer rubrum</i>	Tolerant	28	15	27	24 Aug.-20 Sept.	13	7.2	66
<i>Quercus rubra</i>	Intermediate	20	65	33	19 Aug.-21 Sept.	13	7.4	67
<i>Liriodendron tulipifera</i>	Intolerant	26	30	62	14 July-14 Sept.	14	9.1	71
<i>Populus tremuloides</i>	Very intolerant	15	15	62	30 June-31 Aug.	5	9.8	72
1966								
<i>Fagus grandifolia</i>	Very tolerant	0	81	45	18 July-1 Sept.	12	9.8	72
<i>Liriodendron tulipifera</i>	Intolerant	0	77	45	18 July-1 Sept.	15	9.8	72

* Number of plants taken at each harvest in each treatment.
† The mean of daily minimum and maximum temperatures.

MATERIALS AND METHODS

The species used and their degree of shade tolerance (Baker, 1950) are given in Table 1. In 1965, five species were grown. *Fagus grandifolia* Ehrh., *Acer rubrum* L. and *Quercus rubra* L. were raised from seed, whilst 1-year-old seedlings of *Liriodendron tulipifera* L. and root suckers of *Populus tremuloides* Michx. were used. In 1966 the experiment was repeated, using *Liriodendron tulipifera* and *Fagus grandifolia* only, both raised from seed.

The plants were grown individually in 15 cm nursery cans in a mixture of sand and compost. So that light would remain the main variable controlling growth, the pots were kept well watered. Even so, in the open, the soil dried very quickly in hot weather; consequently, in 1966, the pots were placed in shallow, water-filled trays to maintain an adequate soil moisture content. Liquid fertilizer, supplying approximately 250 mg nitrogen, 125 mg phosphorus and 300 mg potassium per pot, was applied twice during the growing period.

In 1965, the plants raised from seed were transferred to each of four different light treatments, within 2–6 weeks after the seed was sown in the pots (Table 1). Usually at this stage, the plants had opened their cotyledons and one pair of leaves. In 1966, the pots of *F. grandifolia* and *Liriodendron tulipifera* were distributed between the four treatments from the start; therefore all leaves developed in the appropriate light regime.

The four treatments were, full daylight, and three degrees of shade. The latter were obtained using wooden frames, covered with saran shade cloth (Tygan), which transmitted 44, 17, or 3% of full daylight; for further details see Loach (1967). Harvest dates were largely determined by the availability of material, but wherever possible, a lengthy period of adaptation to the shade conditions were allowed before the initial harvest (Table 1).

At first harvest, the length and breadth of each leaf on every plant were measured (including those plants which would later form the second harvest). Leaf areas of the plants actually harvested at this time were determined by cutting out and weighing prints of the leaves made on paper of uniform weight per unit area. The data obtained at first harvest were used solely to establish relationships between (a) total leaf area per plant and the sum of the products of length times breadth for all of its leaves ($\bar{A}$) and (b) plant dry weight and $\bar{A}$. Using these relationships, the dry weight and leaf areas of unharvested plants could be estimated from their leaf measurements. This procedure improves accuracy in determining relative growth rates and net assimilation rates, over the simpler method relying on accurate pairing (McIntyre and Williams, 1949). In the case of *Acer rubrum*, the leaves were too small and numerous for measurement, therefore many paired replications were used (Table 1).

Root suckers of *Populus tremuloides* were pruned to produce more uniform starting material. Stem diameter measurements (by micrometer) were then used in place of leaf measurements for estimating the initial dry weight of the second harvest plants, because the correlation between these two parameters was higher than that between dry weight and leaf measurements. Because of this initial pruning of *P. tremuloides*, direct comparisons of the ratios of leaf, stem and root were not made. Forty plants of this species allowed only a five-fold replication within each treatment; therefore, the derived parameters have a greater uncertainty than those for other species.

Calculation of growth parameters

Relative growth rates (G) were calculated in the usual manner, i.e. $G = (\log_e W_2$

$-\log_e W_1)/t_2 - t_1$, where W_1 and W_2 are oven-dry weights of the whole plants at first harvest (time t_1) and second harvest (t_2). Relative leaf growth rates (G_L) were correspondingly calculated as $(\log_e L_2 - \log_e L_1)/t_2 - t_1$, where L_1 and L_2 are leaf areas per plant at first and second harvest.

The instantaneous net assimilation rate (E) is $(dW/dt)_1/L$ and this is integrated to give a mean value for the whole growth period ($\bar{E}$). This requires information on the relationship between W and L , which was assumed to be of the generalized form, $W = a + bL^n$ (Evans and Hughes, 1962). If $a = 0$, the exponent n is easily calculated as the ratio of relative growth rate to relative leaf growth rate. Plots of W against L for the individual replicates of the first harvest plants showed in the present case, that with one exception, a (the intercept on the W axis) was zero, or negligibly small. n was therefore calculated as G/G_L .

The procedure used subsequently was that recommended by Evans and Hughes (1962, p. 324). Firstly $\bar{E}$ was calculated by the simplest available formula, i.e.

$$\bar{E} = \frac{W_2 - W_1}{L_2 - L_1} \cdot \frac{2}{1}$$

(which assumes that $W = a + bL^2$; Coombe, 1960). This value of $\bar{E}$ was then corrected using the value of n calculated as above, and the correction table given by Evans and Hughes (their Table 1).

In the exceptional case of *P. tremuloides*, where as previously mentioned the correlation between W and L was low, uncorrected values of $\bar{E}$ were used.

Leaf area ratio at final harvest (L_2/W_2), specific leaf area (L_2/L_{W2} , where L_W = leaf weight) and leaf weight ratio (L_{W2}/W_2) were also calculated.

RESULTS

The primary data referring to the plant dry weights and leaf areas at the final harvest, are given in Table 2. The derived growth measures are presented in Figs. 1-6, and the major features of the results reviewed below.

Net assimilation rate ($\bar{E}$)

In full daylight the net assimilation rates were greatest in the 1-year-old plants of the shade-intolerant species, *Populus tremuloides* and *Liriodendron tulipifera*, but these high rates declined rapidly with increasing shade (Fig. 1a). The first-year seedlings of *Quercus rubra*, *Fagus grandifolia*, *Acer rubrum* and *Liriodendron tulipifera* (1966) had considerably lower net assimilation rates in full daylight compared to the year-old plants above, e.g. in *L. tulipifera*, $\bar{E}$ for the first-year seedlings (1966), was only half that of the older plants (1965), despite increased radiation in 1966 (Table 1).

In these young seedlings, $\bar{E}$ was little lower at 44% light than at full light, and in several cases, $\bar{E}$ at 17% was little greater than at 3% light. Although the relationship of $\bar{E}$ and log relative light intensity approached linearity, as found by Blackman and Wilson (1951a), there were some slight departures at the highest and lowest intensities.

Leaf production and related growth measures

The decline in $\bar{E}$ due to shading was partially offset by an increase in the relative size of the photosynthetic system, i.e. an increase in the proportion of leaf area to total plant weight (leaf area ratio, Fig. 2). More of this increase was due to an expansion of the

leaf surface per unit of leaf weight (increased specific leaf area, Fig. 3), than to changes in the proportion of assimilate diverted into leaf production (increased leaf weight ratio, Fig. 4). For example in *L. tulipifera* seedlings (1966), the specific leaf area was nearly three times greater in the deepest shade treatment than in full daylight, but the leaf weight ratio increased by less than half.

One-year-old seedlings of *L. tulipifera* (1965) had lower values of leaf area ratio, specific leaf area and leaf weight ratio, than the first-year seedlings (1966), thus illustrating the usual ontogenetic drift in these parameters. Comparisons between seedlings of *L.*

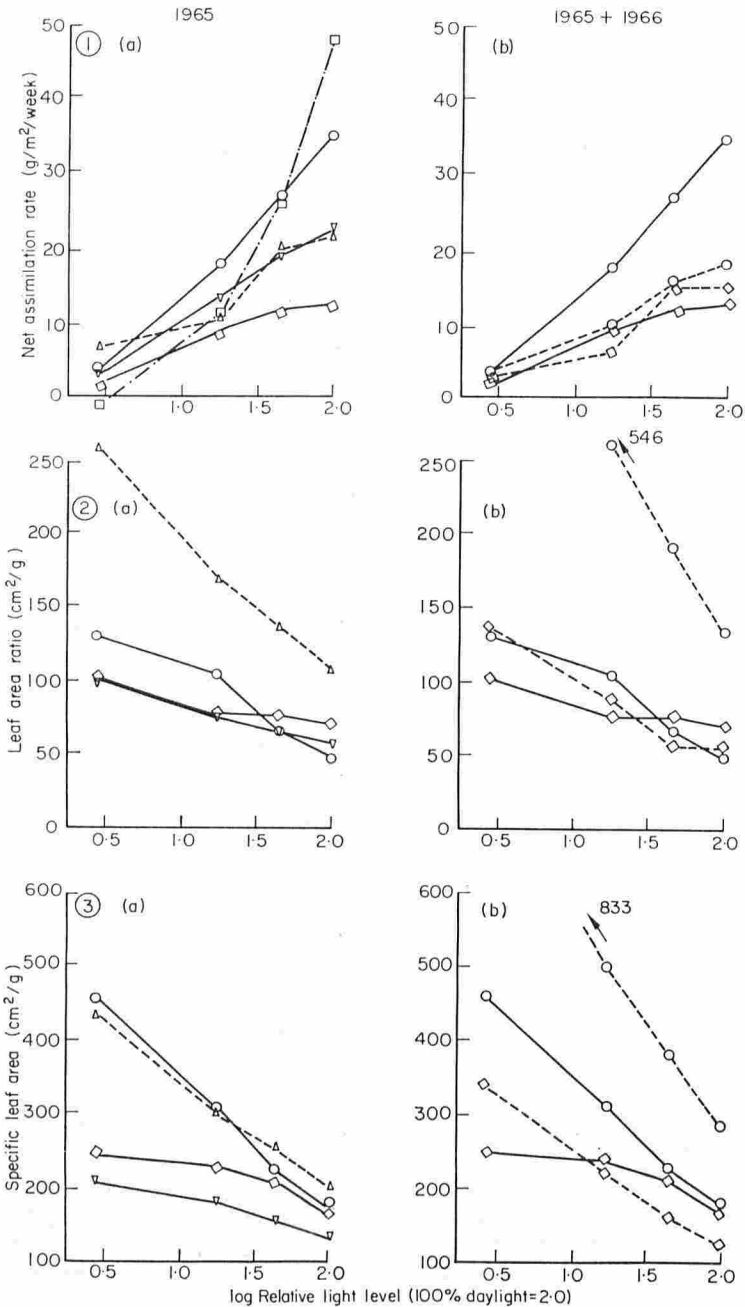
Table 2. *Primary data (mean values per seedling at final harvest)*

Species	Relative light intensity (% daylight)	Dry weight (g)			Total leaf area (cm ²)	No. of leaves
	Root	Stem	Leaf			
1965						
<i>Fagus grandifolia</i>	100	0.46	0.27	0.61	103	6.1
	44	0.51	0.29	0.47	96	5.8
	17	0.39	0.23	0.28	73	4.9
	3	0.15	0.14	0.19	47	3.1
<i>Acer rubrum</i>	100	0.24	0.16	0.45	88	22.6
	44	0.29	0.21	0.59	150	21.1
	17	0.12	0.11	0.30	85	14.2
	3	0.01	0.02	0.04	19	8.1
<i>Quercus rubra</i>	100	2.35	0.68	2.51	274	13.5
	44	2.18	0.49	1.94	289	10.1
	17	1.27	0.39	1.09	180	8.6
	3	0.43	0.15	0.53	99	4.8
<i>Liriodendron tulipifera</i>	100	1.87	0.76	0.99	834	33.6
	44	1.96	0.82	1.14	1293	23.7
	17	1.22	0.69	0.99	1651	23.3
	3	0.64	0.48	0.43	312	16.5
<i>Populus tremuloides</i>	100	2.38	0.70	2.50	275	75.0
	44	2.17	0.50	1.93	303	81.0
	17	1.27	0.38	1.09	195	72.0
	3	0.43	0.16	0.53	114	52.0
1966						
<i>Fagus grandifolia</i>	100	0.33	0.20	0.38	47	6.7
	44	0.33	0.21	0.29	47	4.6
	17	0.19	0.15	0.23	49	4.0
	3	0.05	0.08	0.08	29	2.1
<i>Liriodendron tulipifera</i>	100	0.30	0.13	0.34	95	6.2
	44	0.20	0.12	0.29	112	5.8
	17	0.14	0.09	0.24	115	5.9
	3	0.01	0.01	0.04	32	4.8

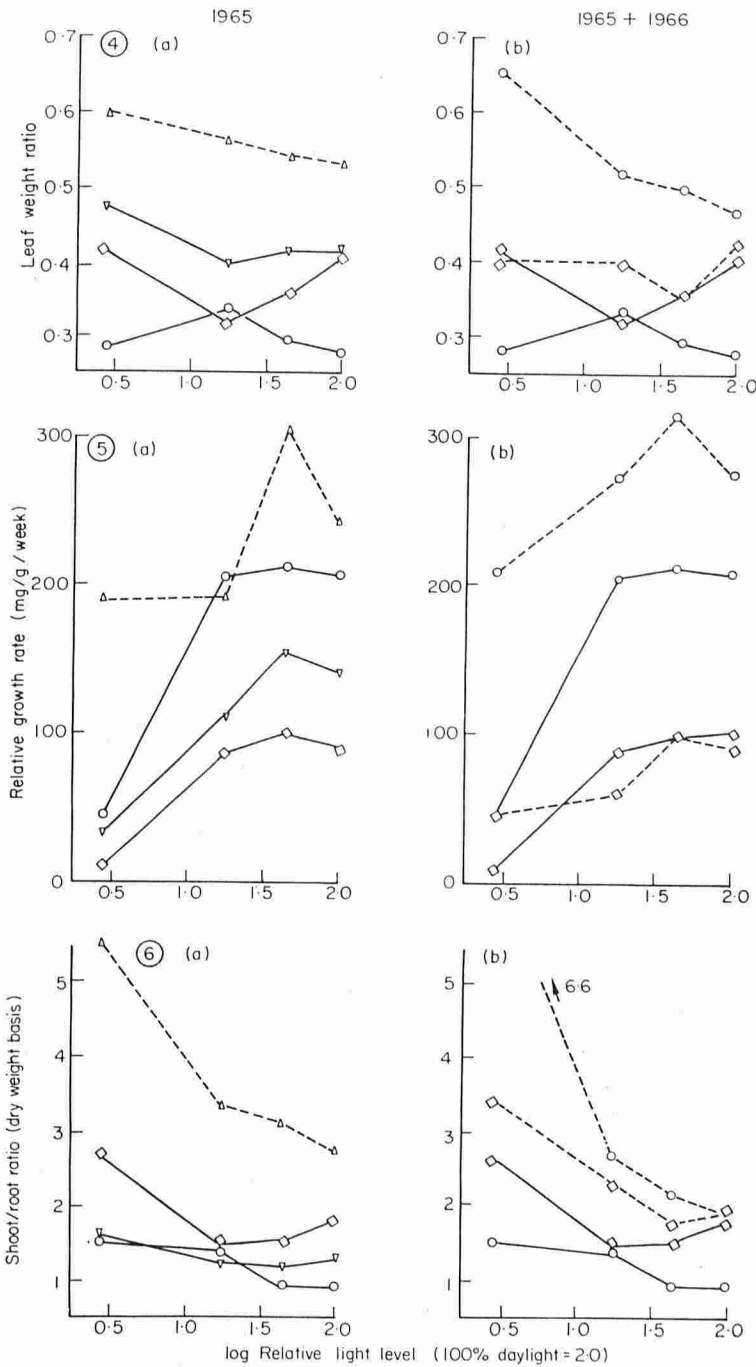
tulipifera (1966) and other species refer to plants of roughly similar age and therefore have most validity.

The relationship between specific leaf area and the logarithm of the relative light intensity (Fig. 3) is approximately linear between full daylight and 17% daylight, and the response to shade over this range can be conveniently measured as the slope of these lines (Table 3). The most responsive species was *L. tulipifera*, the least responsive *Quercus rubra*. Except in the deepest shade, the specific leaf area of *Fagus grandifolia* seedlings grown in 1965 was higher than in 1966, but the response range in 1965 was but a third of that in 1966; the cause of this difference will be discussed later.

The species differed considerably in the rate at which they opened new leaves between harvests (Table 4). In general they produced fewer leaves as shade increased. *F. grandifolia*



Figs. 1-3. The relation between log relative light level (100% daylight = 2.0) and (1) mean net assimilation rate, (2) final leaf area ratio, (3) final specific leaf area, for *Fagus grandifolia* (◇), *Acer rubrum* (Δ), *Quercus rubra* (▽), *Liriodendron tulipifera* (○) and *Populus tremuloides* (□). In all figures (a) gives 1965 data for all species; (b) compares 1965 (—) and 1966 (---) data, for *Fagus grandifolia* and *Liriodendron tulipifera* only.



Figs. 4-6. The relation between log relative light level (100% daylight = 2.0) and (4) final leaf weight ratio, (5) mean relative growth rate and (6) final shoot/root ratio. Symbols as in Figs. 1-3.

produced fewest leaves and *Acer rubrum* the most. Except in the deepest shade, 1-year old plants of *Liriodendron tulipifera* (1965) opened leaves rapidly, but first-year seedling of this species (1966) had a much lower rate of leaf production, which was unaffected by light. The significance of these differences in leaf production, in relation to the calculation of the various growth parameters, will be further discussed in the next section.

Although the changes in leaf weight ratio induced by shade were not large, there were different trends in the various species (Fig. 4). Those species that were most responsive to shade in terms of increased specific leaf area (Table 3), had greater leaf weight ratios in shade than in full daylight, i.e. they diverted relatively more of their assimilate into leaf production when grown at low light intensity. *Quercus rubra* and *Fagus grandifolia* showed relatively little increase in specific leaf area when shaded and reduced their leaf weight ratios in all but deepest shade. Increases at the lowest light intensity probably occurred because assimilate was in very short supply and therefore used near the site of synthesis, in the leaf.

Table 3. *The increase in specific leaf (cm² g⁻¹) per unit decrease in the logarithm of the relative light intensity, between 100% and 17% daylight*

<i>Fagus grandifolia</i> (1965)	91	<i>Liriodendron tulipifera</i> (1965)	178
<i>F. grandifolia</i> (1966)	126	<i>L. tulipifera</i> (1966)	283
<i>Acer rubrum</i>	130	<i>Quercus rubra</i>	68

Shoot : root ratios

Because the proportional contribution of stem weight to total plant dry weight was generally little affected by shade, the shoot : root ratios behaved in an opposite fashion to leaf weight ratios. Thus in *Liriodendron tulipifera* and *Acer rubrum*, roots decreased more than shoots as shade increased, whereas in *Quercus rubra* and *Fagus grandifolia*, shoot : root ratio was little affected, or even decreased in shade, except in 3% daylight (Fig. 6).

Relative growth rate (G)

In nearly every case, the relative growth rate (Fig. 5) was slightly higher in 44% daylight than in full daylight, but significantly so only in *Acer rubrum* and *Liriodendron tulipifera* (1966). This occurred because decline in net assimilation rate over this light range was small, and more than compensated for by the increase in leaf area ratio. *Acer rubrum*, an extreme example, had a much greater relative growth rate in slight shade. In contrast, *Fagus grandifolia* showed little change in $\bar{E}$ or leaf area ratio between full daylight and 44% daylight, and hence a similar relative growth rate at both levels.

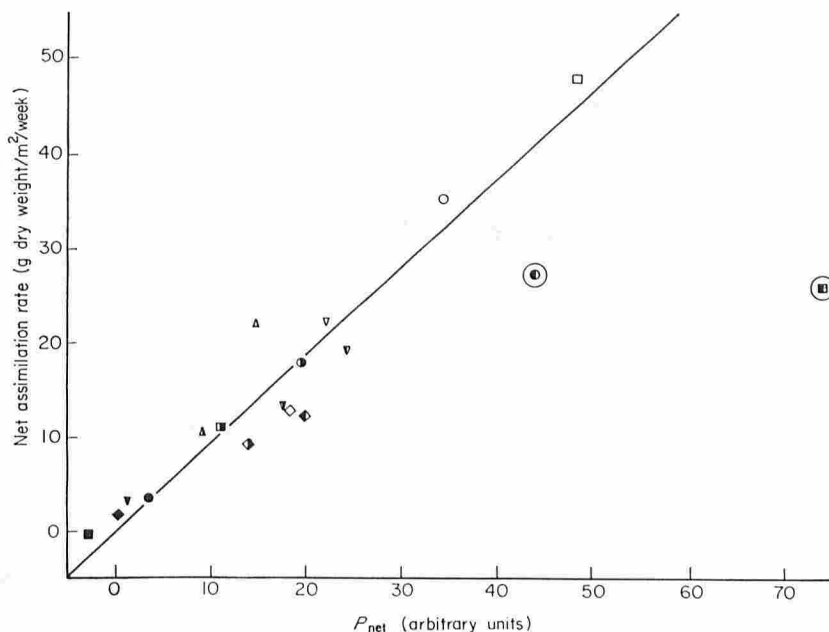
At 17% daylight, the decline in $\bar{E}$ was greater, and the increase in leaf area ratio was seldom sufficient to compensate for it. G was therefore usually lower than in full daylight. Surprisingly, the shade-intolerant *Liriodendron tulipifera* showed least change in G except at the very lowest intensity.

DISCUSSION

The relative ability of different species to tolerate shade may be explained by (1) adaptations involving the efficiency of their photosynthetic system, e.g. low respiration or high photosynthesis rates at low light intensity, or (2) morphological adaptations, involving change in the size of their photosynthetic system. Growth analysis provides a means of

separating the two and assessing their relative contributions. They are discussed in turn below.

Photosynthetic efficiency was measured here as net assimilation rate ($\bar{E}$), i.e. the dry matter gain by the plant between two harvests, expressed on a leaf area basis. $\bar{E}$ has positive (photosynthetic) and negative (respiratory) components, which are most easily examined by making gas exchange measurements on the leaves. Such measurements were previously reported (Loach, 1967; Figs. 1–5). They were made over a short period of time (a matter of minutes) and their reliability in explaining differences in dry matter production over a longer period was assessed by comparing $\bar{E}$ with the net carbon dioxide



Figs. 7. The relation between net assimilation rate ($\bar{E}$) calculated from dry weight increase between harvests, and net carbon dioxide gain per unit leaf area (P_{net}), based on integrated gas exchange measurements (see text). Symbols for species as in Figs. 1–3. Treatments were, 100% daylight ($\circ$), 44% daylight ($\bullet$), 17% daylight ($\bullet$) and 3% daylight ($\bullet$).

gain per unit leaf area (P_{net}), based on gas exchange measurements integrated as follows:

$$P_{\text{net}} = P^{1.0} + 2P^{0.5} - 2R$$

$P^{1.0}$ is the net photosynthesis rate ($\text{mg CO}_2 \text{ dm}^{-2} \text{ hour}^{-1}$) measured in full sunlight, i.e. 10,000 ft-candles (or 146 mW cm^{-2} total short-wave radiation) for plants grown in full daylight, 4400 ft-candles for plants in the 44% daylight treatment, etc. $P^{0.5}$ is the net photosynthesis rate at half sunlight, i.e. 5000 ft-candles for unshaded plants, 2200 ft-candles for plants in 44% daylight, etc. R is the dark respiration rate. The calculation assumes that the plants photosynthesize at rates corresponding to half of full sunlight for 40% of the day, full sunlight for 20% and darkness for the remaining 40%.

$\bar{E}$ includes the respiration of the root and stem, whilst P_{net} is based on photosynthesis and respiration measurements of leaves only. A strict numerical comparison between the two is therefore not possible, and P_{net} is reported in what may be regarded as arbitrary

units (actually $\text{mg CO}_2 \text{ dm}^{-2} \text{ 5 hour}^{-1}$, since it represents the arithmetical sum of five hourly rates).

Considering the obvious approximations involved in P_{net} , and that gas exchange measurements were made on only one leaf per plant (the youngest fully-expanded leaf) the correlation between $\bar{E}$ and P_{net} is remarkably good (Fig. 7). Leaving out the two apparently aberrant points (circled in Fig. 7), the relationship is expressed by: $\bar{E} = 0.93 P_{\text{net}} + 0.24$, with $r^2 = 0.92$. The aberrant points refer to leaves of *Populus tremuloides* and *Liriodendron tulipifera* grown in 44% daylight, which had unusually high rates of photosynthesis (30 and 17 $\text{mg CO}_2 \text{ dm}^{-2} \text{ hour}^{-1}$ respectively). These measurements may have been in error, though agreement between measurements on duplicate leaves was good. Overall, the results suggest that we may have confidence in using the gas exchange measurements to explain differences in $\bar{E}$.

Populus tremuloides grown in 3% daylight had a negative $\bar{E}$ and this was caused by a high respiration rate in darkness (Table 3; Loach, 1967), rather than low photosynthesis rate. Only at the very lowest light intensities (less than about 200 ft-candles or 3 mW cm^{-2}), was its net photosynthesis rate appreciably lower than that of the more tolerant species (Fig. 5; Loach, 1967). Judging from the intercept on the abscissa in Fig. 1(a), the light compensation point for *P. tremuloides* (the point at which photosynthetic gain balances respiratory loss), was about 3.5% daylight.

Intolerance of shade in *Liriodendron tulipifera* cannot be explained on the same basis as in *Populus tremuloides*, because its $\bar{E}$ (and P_{net}) at 3% daylight was positive, and greater than in the shade-tolerant species (Fig. 7). The relatively steep terminal slope in Fig. 1(a) suggests that it may nevertheless have a higher compensation point than the more tolerant species, but it is doubtful that the small difference involved could be of any significance. Fig. 1(b) suggests that younger plants of this species (1966) may have a lower light compensation point and be more tolerant of shade than older plants (1965).

In Fig. 1, net assimilation rates are plotted against the logarithm of relative light intensity. Blackman and Wilson (1951a) found this relationship to be linear between 12% and 100% daylight, but later, when intensities as low as 5.5% daylight were included in their experiments (Blackman and Black, 1959), it was seen to be curvilinear, i.e. the slope changed and $\bar{E}$ fell less rapidly at this lowest intensity. Similar departures from linearity were noted here in the shade intolerant *P. tremuloides* and the tolerant *Acer rubrum* and *Fagus grandifolia* (1966). Thus both tolerant and intolerant species show this measure of adaptability to low light levels.

Deviations from linearity in the curves of $\bar{E}$ versus log relative light intensity have also been noted at the highest light intensities. Reduced net assimilation in full sunlight has often been found in tree species (Shirley, 1929; Polster, 1950; Jarvis, 1964; and others) and has been variously ascribed to light damage, water stress and stomatal closure or to nutrient deficiency in sun-grown leaves. Leaves of *F. grandifolia* grown here in full daylight were yellowed, which could indicate light damage or nitrogen deficiency. Seedlings of *Acer rubrum* inclined their laminae at right angles to the petioles when grown in full sunlight (see Grime, 1966; Plate 2), which would serve to promote water economy, but at the same time would reduce photosynthesis. Gas exchange measurements showed that leaves of all five species grown in 44% daylight had higher rates of photosynthesis than those grown in full daylight. However, the intolerant *Populus tremuloides* and *Liriodendron tulipifera* showed little evidence of reduced $\bar{E}$ in full daylight (Fig. 1).

To summarize; the negative net assimilation rate of *Populus tremuloides* at low light

intensities, caused by a high respiration rate, accounts for its intolerance of shade. Tolerant species maintained a positive E at low light intensities, but so did the intolerant *Liriodendron tulipifera*, and another explanation must be found for its apparent intolerance of shade.

Figs. 2–4 illustrate the changes in size of the photosynthetic system. The leaf area per unit plant weight (leaf area ratio) increased as light intensity decreased (Fig. 2). Blackman and Wilson (1951b) found that the relationship between leaf area ratio and the logarithm of the relative light intensity was linear, and used the slope of the line as a measure of the sensitivity of the leaf area ratio to shading, i.e. the plasticity of the plant to changes in light intensity. Working with herbs, they found that shade-tolerant species were more plastic in this respect than intolerant species. This led them to define a shade plant as one in which 'a reduction of the light intensity causes a rapid rise in the leaf area ratio from an initial low value in full daylight'.

In the present case a rough linearity holds (Fig. 2), but the steepest slope was found in the shade-intolerant species *L. tulipifera*, whether expressed in absolute units or relative to the leaf area ratio in full daylight (as preferred by Blackman and Wilson). *Fagus grandifolia* (1965), a tolerant species, showed least increase in leaf area ratio when shaded, although plants raised from the start in shade (1966, Fig. 2b), showed a greater increase (see below).

As already mentioned, the leaf area ratio can be further analysed as the product of two components, viz. leaf weight ratio (leaf weight/plant weight), and specific leaf area (leaf area/leaf weight). The relatively small variations in leaf weight ratio shown in Fig. 4, were seldom significant, but within these limits of reliability they do not appear to follow previously reported patterns. Kuroiwa *et al.* (1964) noted that shade-intolerant herbaceous species put less and less of their assimilate into leaf production as the level of shade increased, whereas the tolerant species *Cryptotaenia canadensis* var. *japonica*, invested relatively more assimilate in leaf tissue when shaded. Here, the shade-tolerant *Fagus grandifolia* and *Quercus rubra* had lower leaf weight ratios in 44% and 17% daylight than in full daylight, whereas in the intolerant *Liriodendron tulipifera* (1966) the leaf weight ratio increased with shade (Fig. 4).

It should be noted that specific leaf area and leaf area ratio as expressed here refer to the whole plant. Their value is therefore influenced not only by the adaptability of leaves already in existence at the first harvest (which may be limited for mature leaves), but also by the rate at which new, and presumably more adaptable leaves are produced. Assuming that new leaves are more adaptable than older leaves, then a species with a high rate of leaf production has a greater potential for increasing its specific leaf area than one in which few new leaves are produced.

The rate of leaf production in *L. tulipifera* (1966) was, however, of the same order as that for *Fagus grandifolia* (except in 3% daylight, Table 4), so that the greater increase in specific leaf area by the intolerant species (Table 3 and Fig. 3), was due to a greater expansion of the leaf surface and not to the production of more young leaves. In 1-year-old seedlings of *Liriodendron tulipifera* (1965), the total leaf area of plants raised in 17% daylight was twice that of open-grown plants (Table 2), and the individual leaves were approximately three times as large.

Comparison of *Fagus grandifolia* in the 2 years illustrates the greater adaptability of plants raised from the start in the shade treatments (1965), as compared with those transferred to the shade after the first pair of leaves had opened (1965). Differences in seed source for plants grown in different years may also have contributed to the different

responses. Radiation levels were similar (Table 1) in both years, and could not account for the difference.

Acer rubrum showed the second greatest increase in specific leaf area when shaded. This is perhaps partly a result of the low overall level of radiation during the period between harvests in this instance (Table 1), but also *A. rubrum* had the greatest rate of leaf production of all species (Table 5), which of itself confers a high potential for increasing the specific leaf area of the plant as a whole, as explained above.

In summary; the reduction in $\bar{E}$ in shade was offset by an increase in the ratio of leaf area to plant weight, but the magnitude of this response in different species, does not explain their relative ability to tolerate shade. In fact, the shade-intolerant *Liriodendron tulipifera*, showed greatest increase in leaf area ratio.

The combined effect of changes in $\bar{E}$ and leaf area ratio is indicated by their product, the relative growth rate (Fig. 5). *L. tulipifera* maintained its high relative growth rate (G) down to 17% daylight, and even in 3% daylight its G matched that of more tolerant species. In *Fagus grandifolia* (1966) and *Quercus rubra*, G was in fact slightly, but significantly, reduced in 17% daylight.

Table 4. Rates of leaf production between harvests (number of new leaves produced per week)

Species	Relative light intensity (% daylight)			
	100	44	17	3
<i>Fagus grandifolia</i> (1965)	0.2	0.3	0.2	0.0
<i>F. grandifolia</i> (1966)	0.3	0.1	0.2	0.0
<i>Acer rubrum</i>	2.9	2.5	0.9	0.6
<i>Quercus rubra</i>	0.7	0.3	0.2	0.0
<i>Liriodendron tulipifera</i> (1965)	1.8	1.1	0.8	0.0
<i>L. tulipifera</i> (1966)	0.3	0.3	0.3	0.3

Only four species were examined here and it is clearly impossible to generalize on this narrow basis. However, evidence from other sources also suggests that shade tolerant woody species are not especially characterized by an exceptional ability to increase their specific leaf area and leaf area ratio in shade conditions. Jackson (1967) measured blade thicknesses of sun and shade leaves of twenty-one tree species. The average thicknesses of shade leaves, expressed as a percentage of sun leaf thicknesses, were 86, 77 and 60% for species in tolerant, intermediate and intolerant categories respectively, i.e. intolerant species showed the greatest reduction in thickness when shaded, and presumably therefore, the greatest increase in specific leaf area. Whether this would result in an increased leaf area ratio for the plant as a whole growing in shade remains uncertain, because the leaf weight ratio could be disproportionately reduced by shade in the intolerant species, as the data of Kuroiwa *et al.* (1964) suggested. Further work is needed before generalizations are possible for seedling plants.

From the point of view of light capture in shade conditions, a high specific leaf area cannot be other than advantageous, so that it is difficult to understand why leaves of shade-intolerant species should be more plastic in this respect. There may well be disadvantages concomitant with high specific leaf area however. For example, very thin leaves would lose water rapidly and this may be particularly serious in a species such as *Liriodendron tulipifera* which is particularly susceptible to droughting (Smith, 1963; Stephens, 1965). In this case, the increased specific leaf area is accompanied by an increase in the shoot : root ratio (Fig. 6b), so that water losses are probably aggravated by reduced uptake. Shade leaves of this species were also very susceptible to fungal

attack, possibly because energy reserves were dissipated in expansion of the surface, or because the development of very thin walls facilitated entry of the pathogen.

Why intolerant tree species should show greatest plasticity in response to shade, whilst in herbs, the reverse is apparently the case, is also puzzling. However, intolerant tree seedlings in many cases occupy the same low, shady stratum as many herbaceous species, e.g. in the ground flora of a forest clearing or in grassland. A large leaf area is a desirable asset in enabling the plant to outgrow its competitors quickly and emerge from the shade stratum, even though it may be bought at the expense of short-term risk, such as that of desiccation, should drought occur. Shade-tolerant tree seedlings on the other hand, growing in the taller shade stratum of a temperate woodland, cannot easily escape shade by an initial burst of growth, and slow steady growth and conservation of reserves is then of prime importance. This distinction between shade avoidance by rapid initial height growth, and shade tolerance, where the emphasis is on conservation of reserves, has already been made by Grime (1966). Newhouse and Madgwick (1968) have also found that in tree species, rapid growth is associated with intolerance of shade.

Many authors have noted that the most tolerant tree species are frequently characterized by the maintenance of a uniform pattern of growth and metabolism over a wide range of light intensities, e.g. Baker (1945) in *Pseudotsuga menziesii*, Logan (1965) in *Acer saccharum*, Bordeau and Laverick (1958) in *Tsuga canadensis*, etc. In the present case, the shade-tolerant species *Fagus grandifolia* and *Quercus rubra* were least influenced in their pattern of growth by shade and their metabolism too was relatively unaffected (Loach, 1967). *Acer rubrum*, also tolerant, proved an exception (Fig. 6a), though the respiration rate of its leaves was not increased by shade (Loach, 1967; Table 3).

We can now return to the initial question as to whether differences in shade tolerance between species lie primarily with differences in the size or the efficiency of the photosynthetic system in shade-grown plants. Apparently both play a role, but neither clearly differentiates between tolerant and intolerant species. Thus intolerance in *Populus tremuloides* can be attributed to its high respiration rate and consequent low assimilatory efficiency in shade conditions, but *Liriodendron tulipifera*, also intolerant, was quite as efficient as tolerant species in this respect. In *Acer rubrum*, a large increase in the size of the photosynthetic system enabled it to maintain a high relative growth rate in shade; *Liriodendron tulipifera*, however, showed an even greater increase in leaf area ratio. Whilst both species are apparently capable of maintaining a satisfactory carbohydrate economy in shade, difficulties with water economy may militate against the long-term survival of *L. tulipifera* in shade.

We must conclude with Baker (1945) that no one principle explains differences in shade tolerance between species, and there is 'a tendency to overstress the common similarities of tolerant trees and intolerant trees', whereas 'sharp individual differences are shown'.

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